DAY 02 | HANDS-ON ACTIVITIES

Functions, Data Structures and Files: Problem Statements

Seven problems from warm-up to stretch. Solve them in the Day 2 notebook or in a .py file in VS Code.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Area Helpers

Level: Warm-up | **Concepts:** def, parameters, return

Write three functions: rectangle_area(l, w), circle_area(r) and triangle_area(b, h). Call each and print the result rounded to 2 decimals.

Sample run / expected result

```
rectangle_area(4, 5)  -> 20
circle_area(3)       -> 28.27
triangle_area(6, 4)  -> 12.0
```

Hints

- Use 3.14159 for pi, or import math and use math.pi.

Problem 2: Shopping List Manager

Level: Easy | **Concepts:** list methods, loops

Start with an empty list. Add five items, remove one, sort the list, then print each item with its position number starting from 1.

Sample run / expected result

```
1. bread
2. eggs
3. milk
4. rice
```

Hints

- enumerate(items, start=1) gives position and item together.

Problem 3: Remove Duplicates

Level: Easy | **Concepts:** set, list

Given emails = ["a@x.com", "b@x.com", "a@x.com", "c@x.com", "b@x.com"], print how many unique emails there are and print them sorted.

Sample run / expected result

```
Unique: 3
['a@x.com', 'b@x.com', 'c@x.com']
```

Hints

- set() removes duplicates. sorted() returns a list.

Problem 4: Word Counter

Level: Medium | **Concepts:** dict, string split, functions

Write count_words(text) that returns a dict of each word and how often it appears (ignore case). Print the three most common words.

Sample run / expected result

```
count_words("the cat and the hat and the bat")
{'the': 3, 'cat': 1, 'and': 2, 'hat': 1, 'bat': 1}
Top 3: the, and, cat
```

Hints

- text.lower().split() gives a list of words.
- sorted(d.items(), key=lambda kv: kv[1], reverse=True)

Problem 5: Student Report

Level: Medium | **Concepts:** list of dicts, functions

Store five students as a list of dicts with name and marks. Write functions average(students) and topper(students). Print the class average and the topper's name.

Sample run / expected result

```
Class average: 74.4
Topper: Priya (92)
```

Hints

- max(students, key=lambda s: s["marks"])

Problem 6: Expenses to CSV

Level: Challenging | **Concepts:** csv module, dicts

Write a list of expenses (date, category, amount) to expenses.csv. Then read it back and print the total spent per category.

Sample run / expected result

```
Food: 1450.0
Travel: 800.0
Bills: 2200.0
```

Hints

- Amounts read from CSV are text, so cast with float().
- Use a dict to accumulate totals per category.

Problem 7: Contact Book with JSON

Level: Stretch | **Concepts:** functions, dict, json, files

Build add_contact, find_contact and save_contacts / load_contacts functions. Contacts are stored in contacts.json so they survive restarts.

Sample run / expected result

```
add_contact("Mei", "98450 11111")
save_contacts()
# restart
load_contacts()
find_contact("Mei") -> 98450 11111
```

Hints

- Load returns an empty dict if the file does not exist yet.
- json.dump(contacts, f, indent=2)

Ganbatte (頑張って): do your best